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(54) **WIDE-ANGLE IMAGE CAPTURING LENS ASSEMBLY**

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See application file for complete search history.

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G02B 9/58	(2006.01)
G02B 13/00	(2006.01)
G02B 13/04	(2006.01)

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CPC ... **G02B 9/34** (2013.01); **G02B 9/58** (2013.01);
G02B 13/18 (2013.01); **G02B 13/004**
(2013.01); **G02B 13/04** (2013.01)
USPC **359/715**; 359/781; 359/782

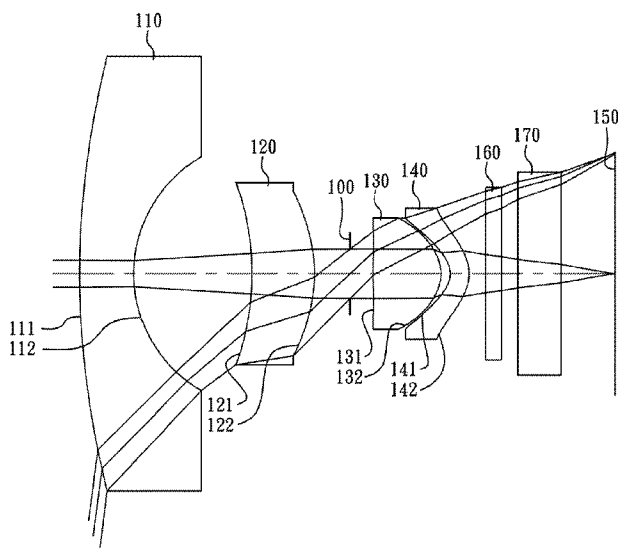
(58) **Field of Classification Search**

CPC G02B 9/34; G02B 9/58; G02B 13/004;
G02B 13/18

ABSTRACT

A wide-angle image capturing lens assembly includes four lens elements with refractive power, in order from an object side to an image side, a first lens element, a second lens element, a third lens element and a fourth lens element and each of the first through fourth lens elements is single and non-cemented. The first lens element with negative refractive power has a concave image-side surface. The second lens element with refractive power has a concave object-side surface and a convex image-side surface. The third lens element with positive refractive power has a convex image-side surface. The fourth lens element with negative refractive power has a concave object-side surface and a convex image-side surface, wherein both of the object-side surface and the image-side surface of the fourth lens element are aspheric.

21 Claims, 12 Drawing Sheets



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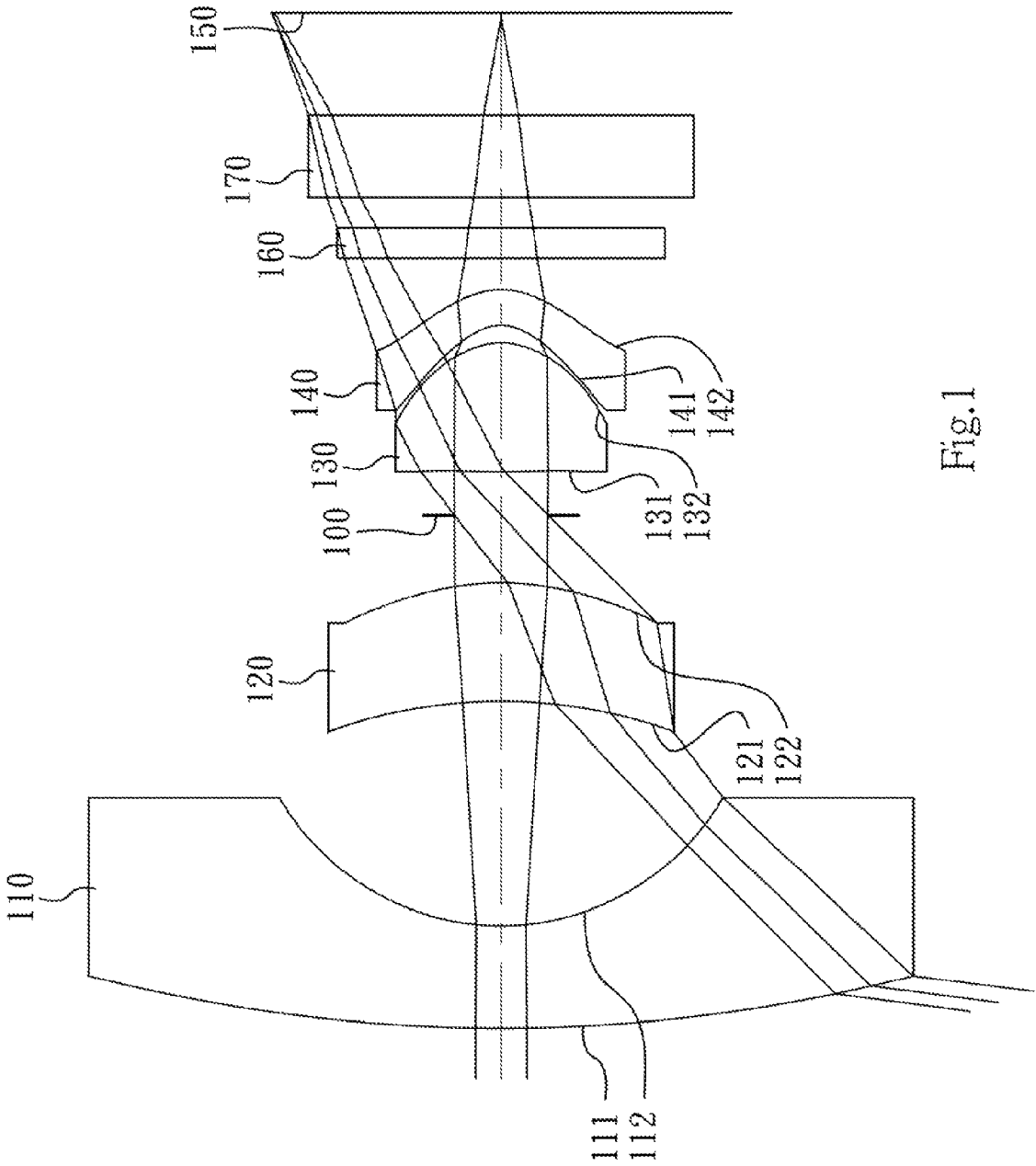


Fig.1

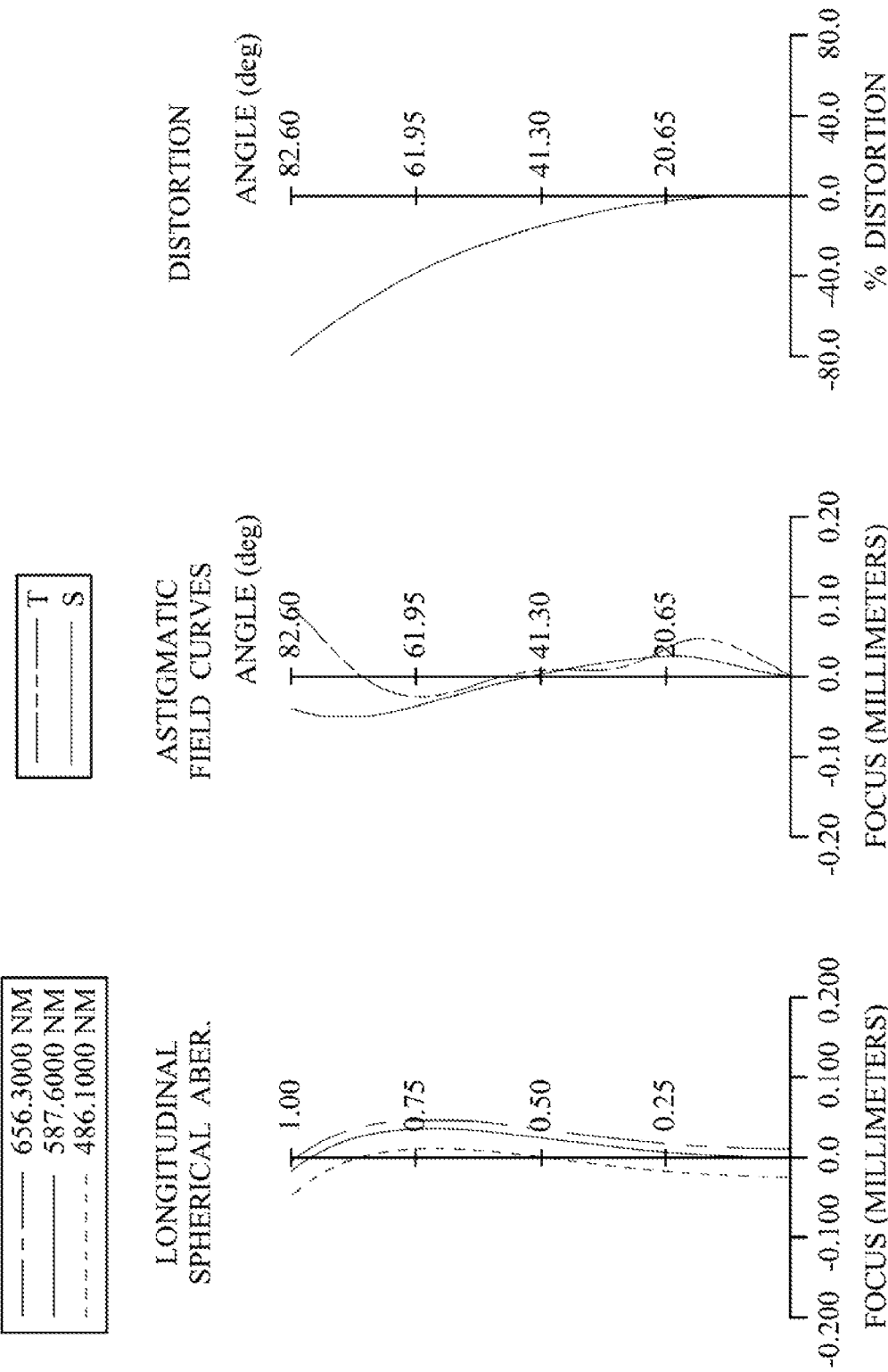


Fig.2

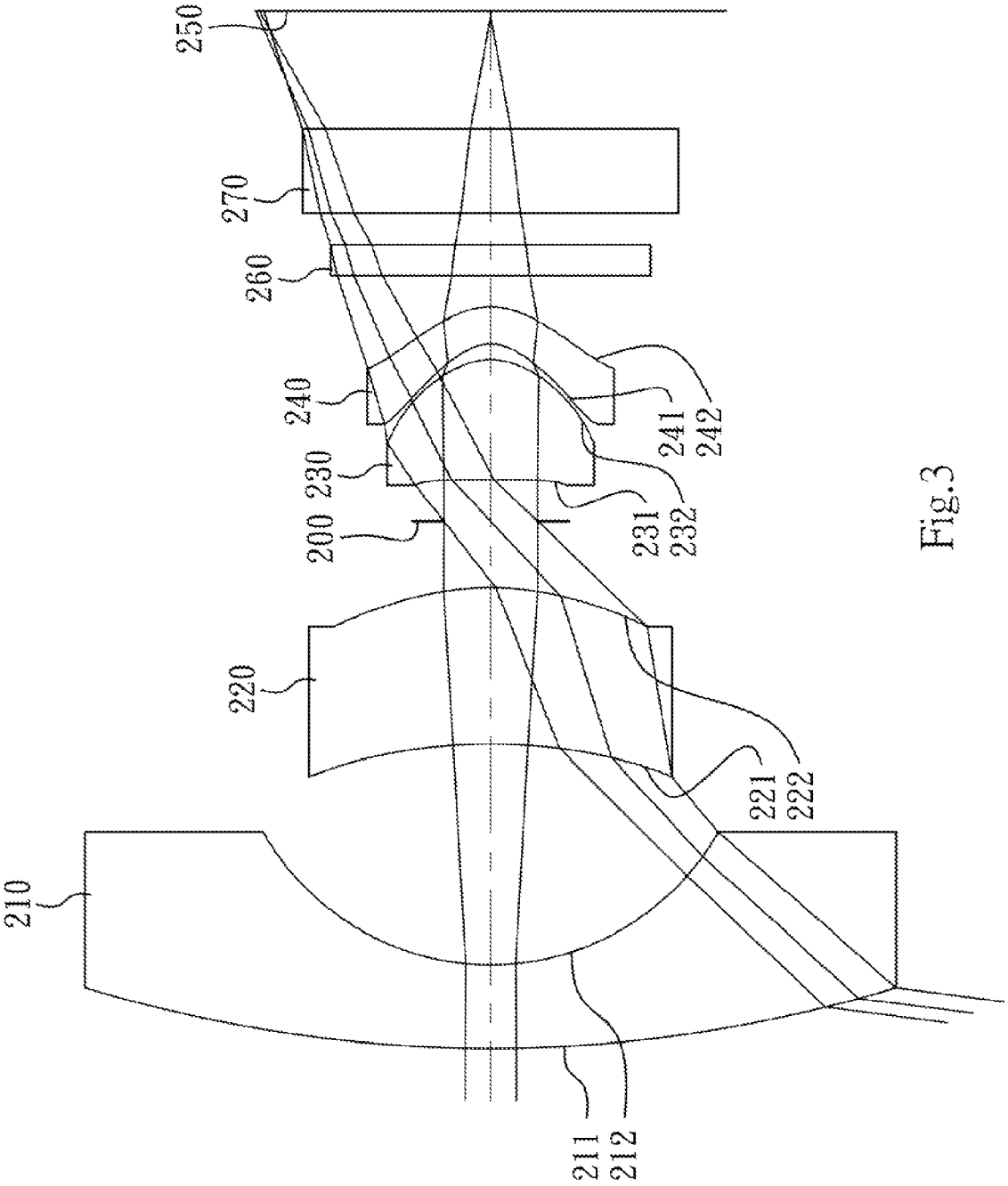
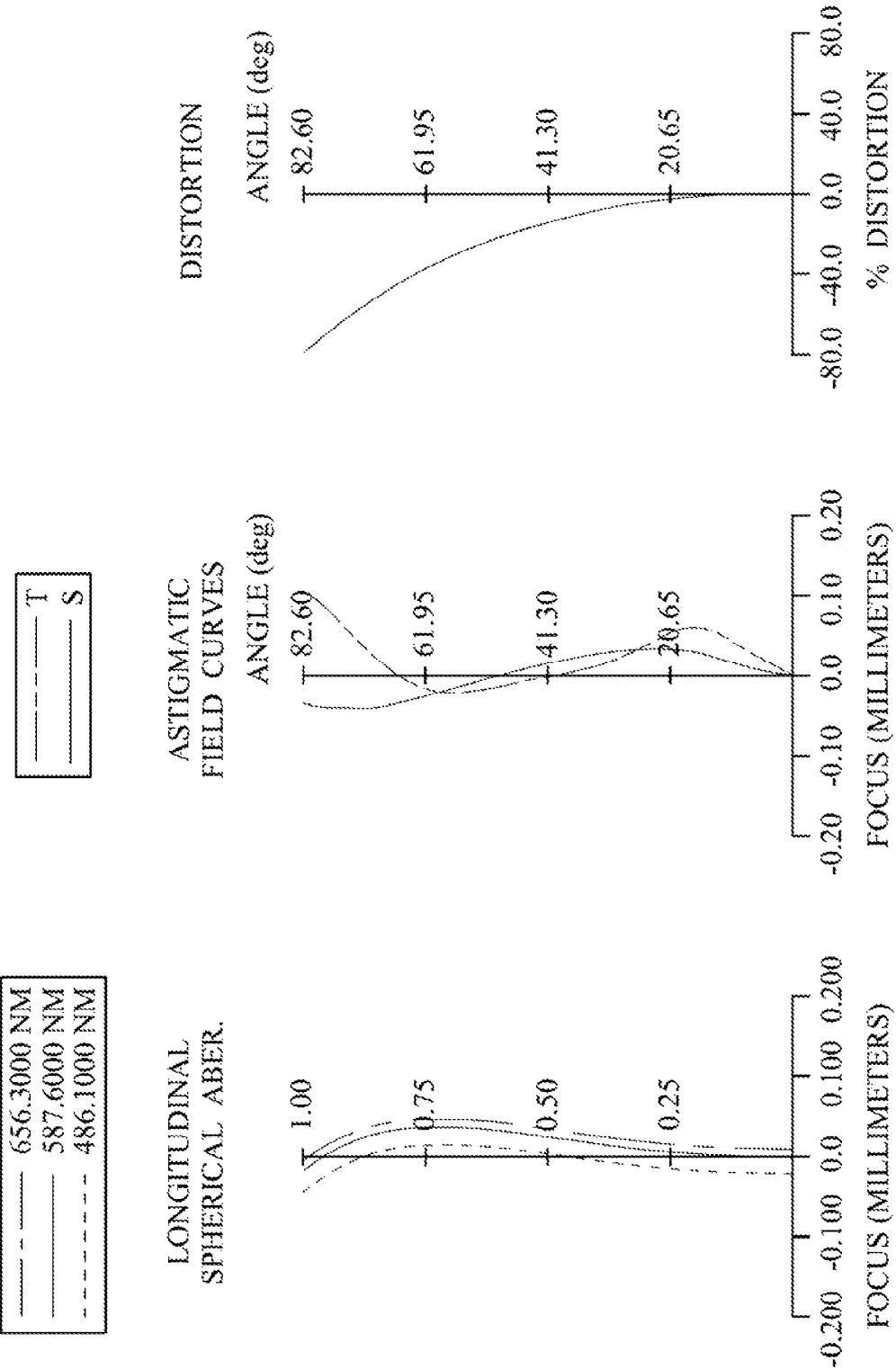


Fig.3



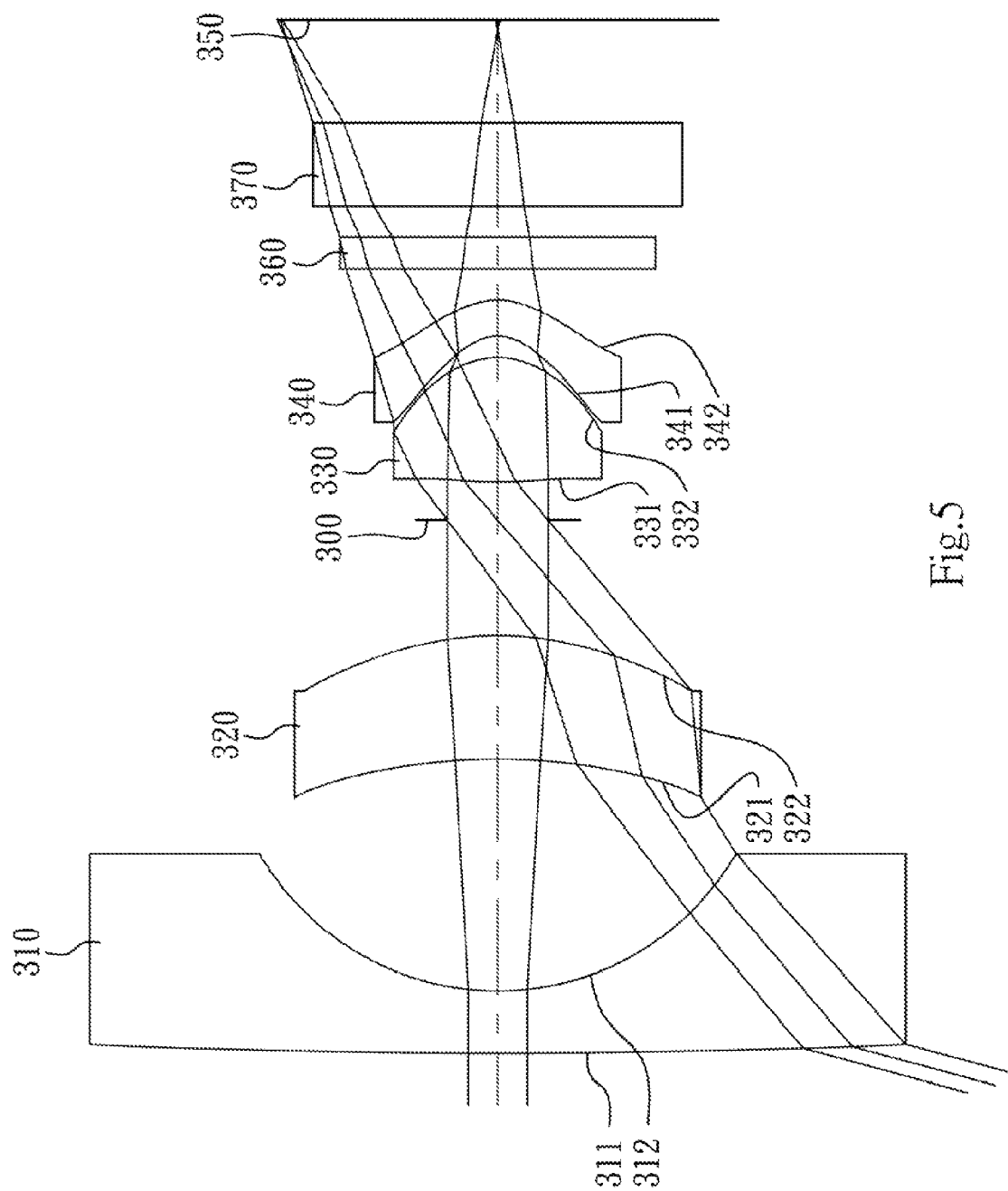


Fig. 5

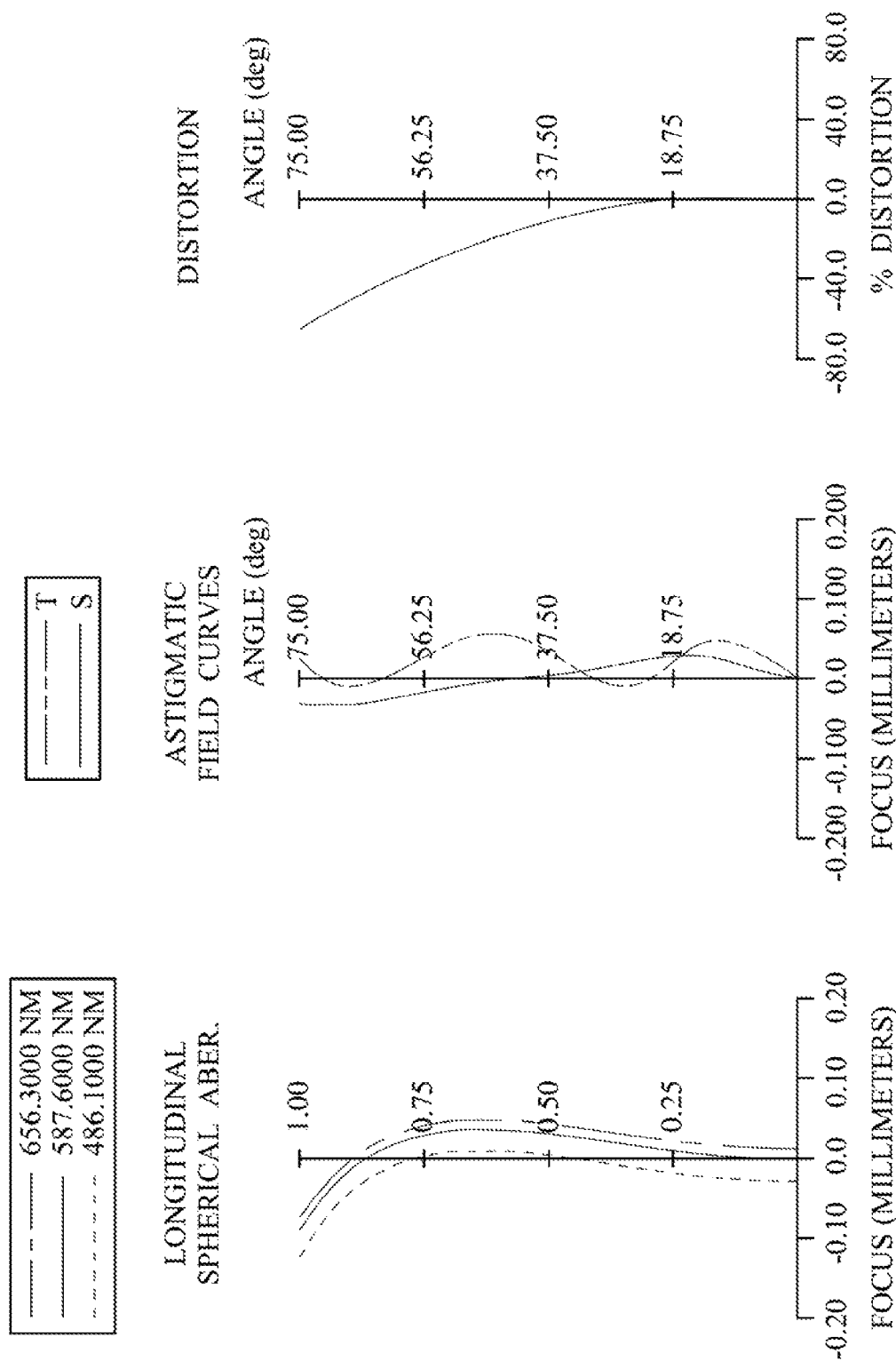


Fig.6

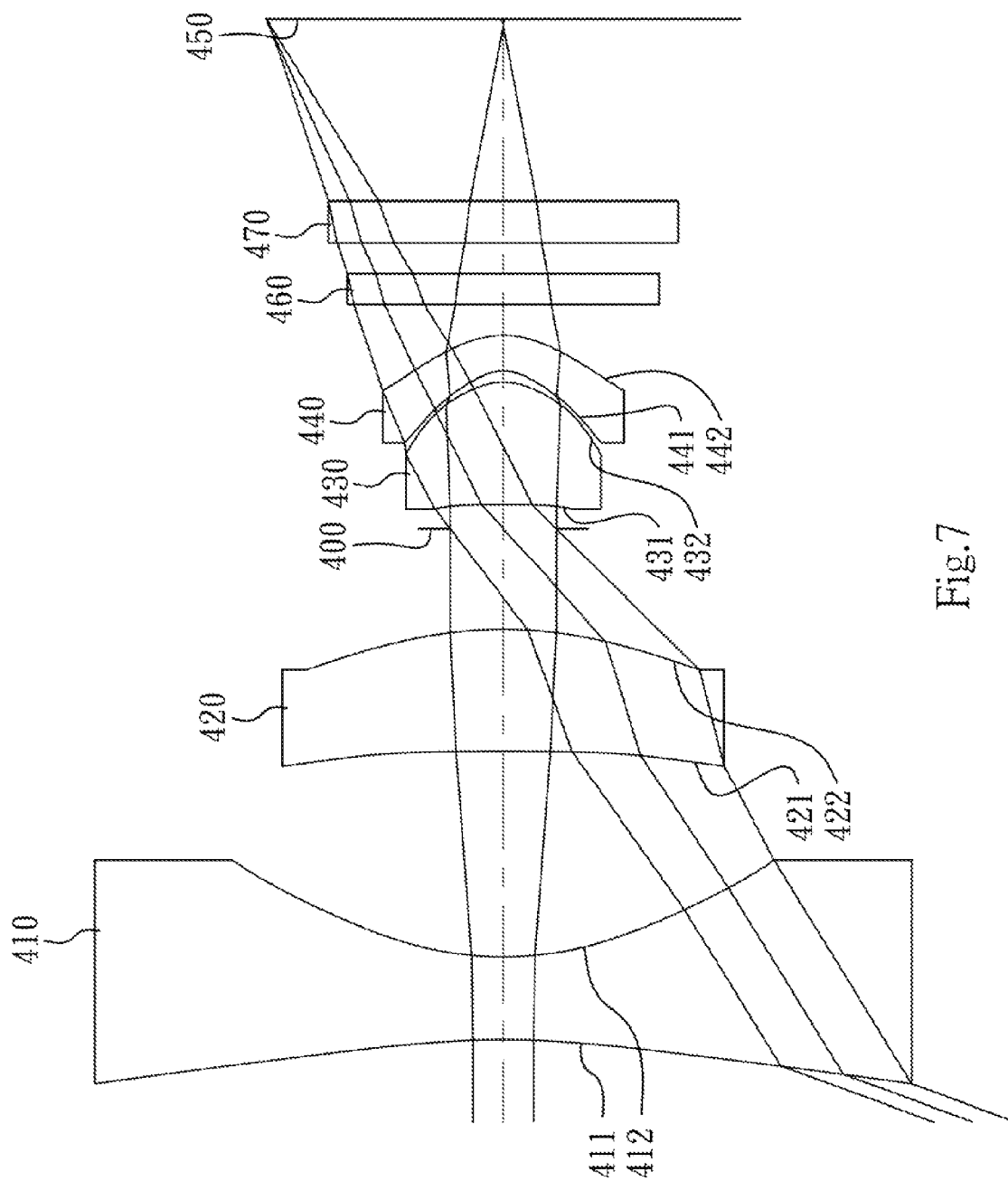


Fig. 7

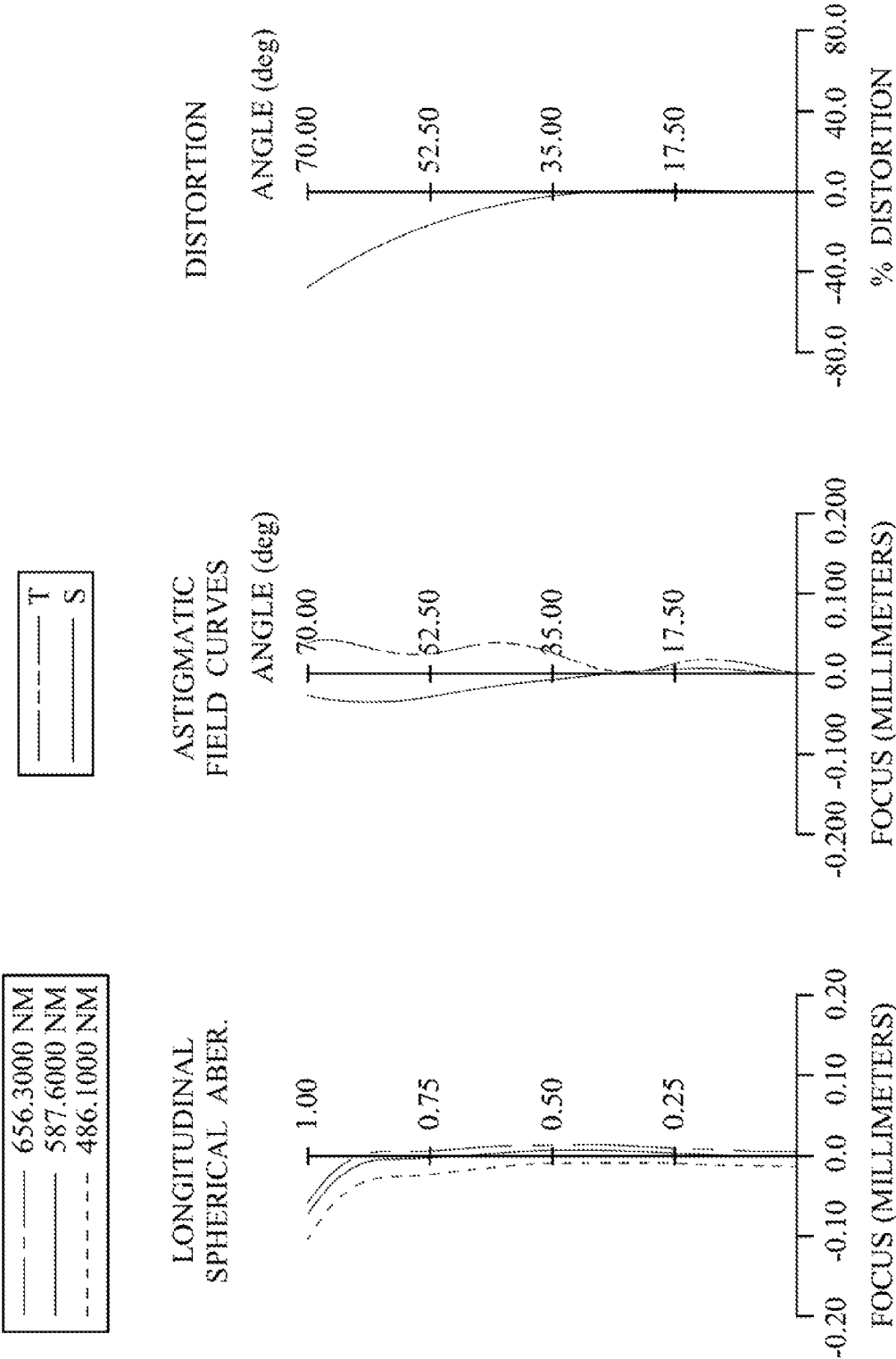


Fig.8

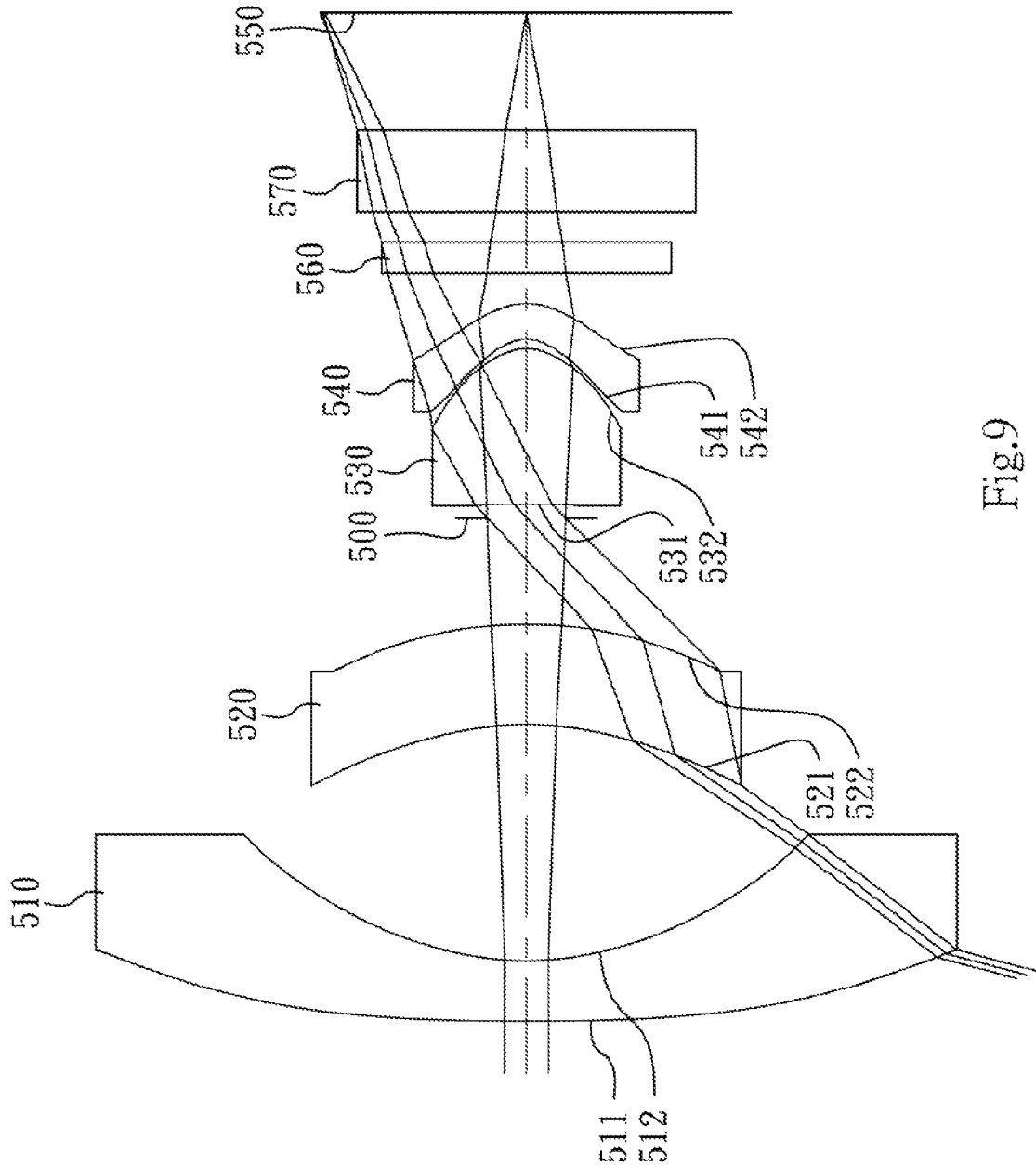
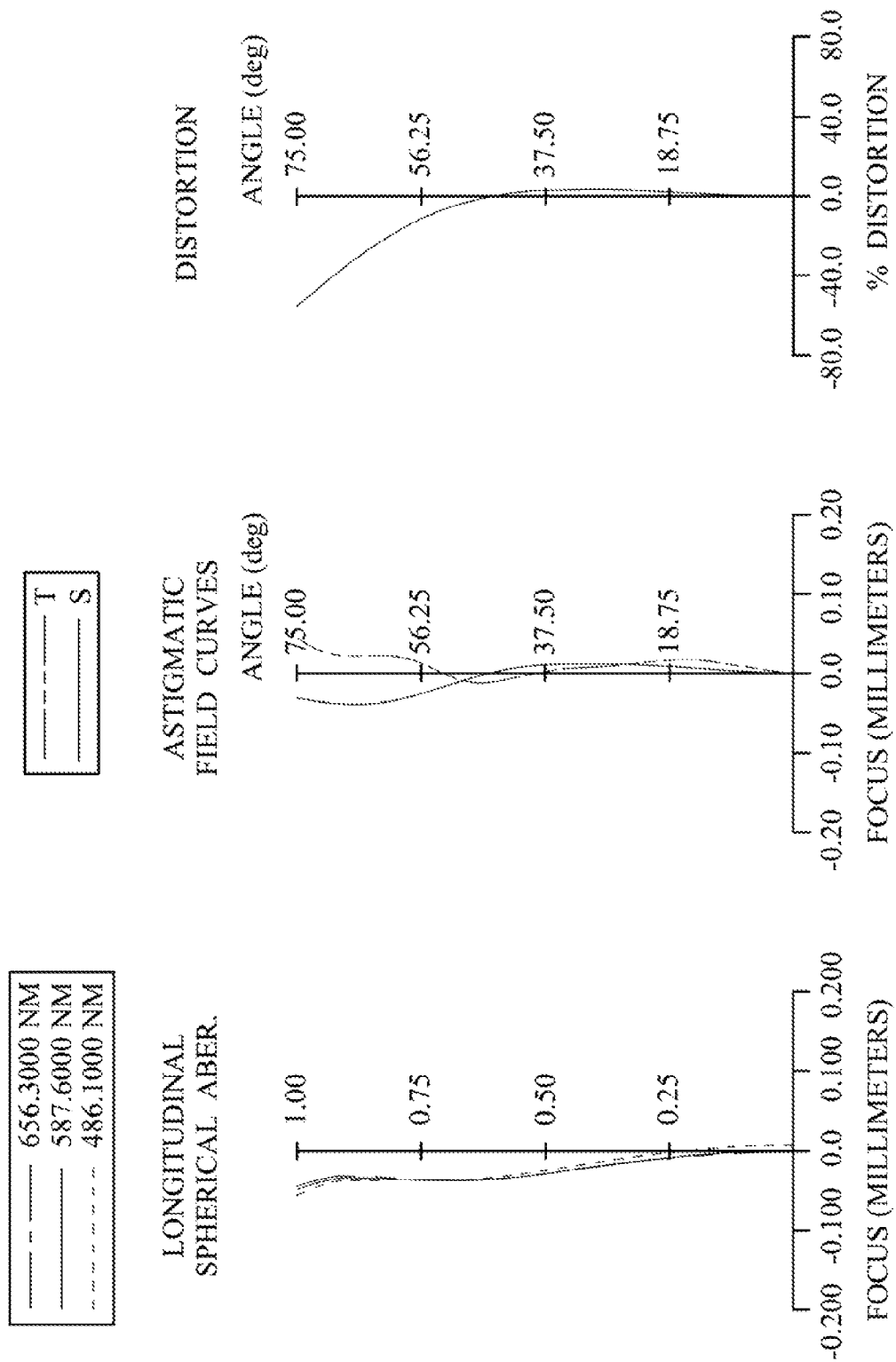


Fig.9



10.0000

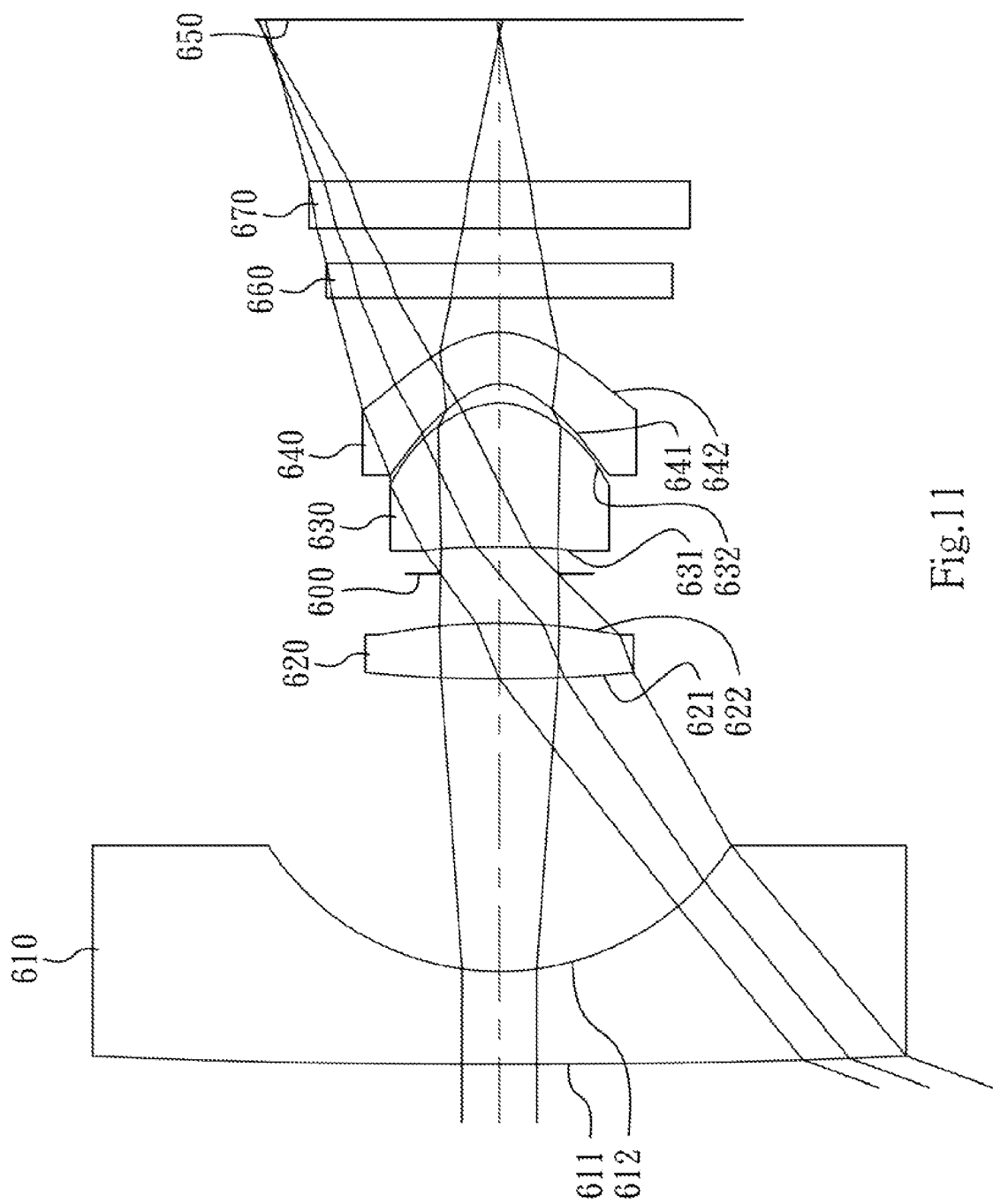


Fig.11

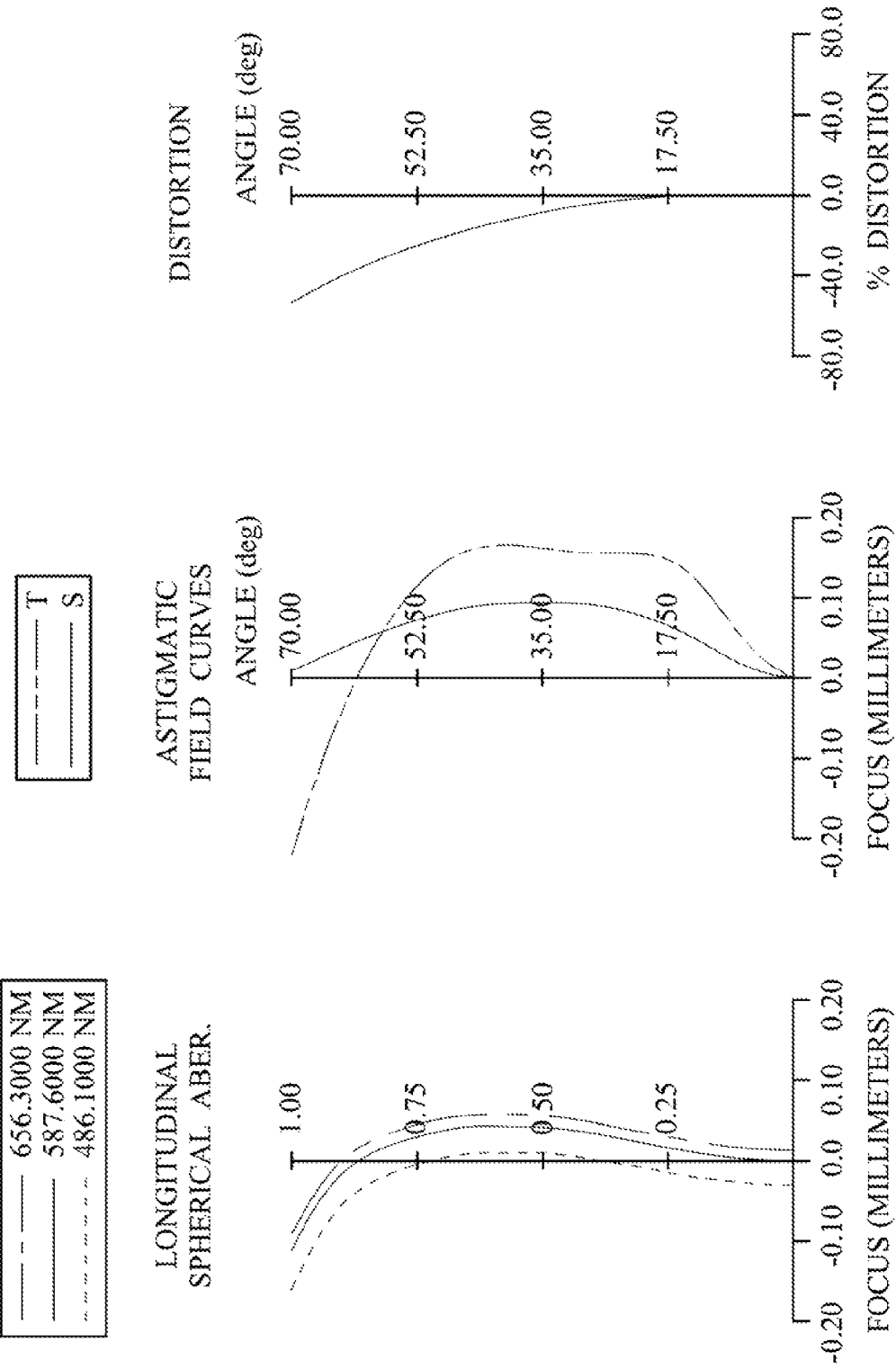


Fig.12

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WIDE-ANGLE IMAGE CAPTURING LENS ASSEMBLY

RELATED APPLICATIONS

The application claims priority to Taiwan Application Serial Number 102105430, filed on Feb. 8, 2013, which is herein incorporated by reference.

BACKGROUND

1. Technical Field

The present disclosure relates to a wide-angle image capturing lens assembly. More particularly, the present disclosure relates to a compact wide-angle image capturing lens assembly applicable to electronic products.

2. Description of Related Art

In recent years, with the popularity of mobile products having camera functionalities, the demand of miniaturized optical lens systems has been increasing. The sensor of a conventional photographing camera is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical lens systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical lens systems featuring better image quality.

A conventional compact optical lens system employed in a portable electronic product mainly adopts a four-element lens structure such as the one disclosed in the U.S. Pat. No. 7,365,920. The positive first lens element and the negative second lens element of the four-element lens structure are attached to each other to form a doublet lens. However, during the cementing process, the doublet lens tends to be decentered. Moreover, this doublet lens might be tilted if the glue is not applied evenly. The degrees of freedom of these two cementing surfaces might also be limited. Therefore, the doublet lens is hard to be applied to the electronic products featuring superior image quality since the manufacturing processes of the doublet lens makes it difficult to improve image quality.

There is another conventional optical lens systems with four-element lens structure such as the one disclosed in the U.S. Pat. No. 8,014,080. Although its first lens element with negative refractive power is favorable for enlarging the field of view, it still cannot satisfy the requirement of present automotive optical lens systems. Furthermore, its surface design of the lens elements is not favorable for correcting the aberration; it is thereby not suitable for the present automotive optical lens systems featuring superior image quality.

SUMMARY

According to one aspect of the present disclosure, a wide-angle image capturing lens assembly includes four lens elements with refractive power, in order from an object side to an image side, a first lens element, a second lens element, a third lens element and a fourth lens element. The first lens element with negative refractive power has a concave image-side surface. The second lens element with refractive power has a concave object-side surface and a convex image-side surface. The third lens element with positive refractive power has a convex image-side surface. The fourth lens element with negative refractive power has a concave object-side surface and a convex image-side surface, wherein both of the object-side surface and the image-side surface of the fourth lens element are aspheric. Each of the first through fourth lens

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elements is a single and non-cemented lens element with refractive power. When a curvature radius of an object-side surface of the third lens element is R5, a curvature radius of the image-side surface of the third lens element is R6, and a curvature radius of the object-side surface of the fourth lens element is R7, the following relationships are satisfied:

$$0.6 < (R5+R6)/(R5-R6) < 2.0; \text{ and}$$

$$0.30 < R7/R6 < 0.74.$$

According to another aspect of the present disclosure, a wide-angle image capturing lens assembly includes four lens elements with refractive power, in order from an object side to an image side, a first lens element, a second lens element, a third lens element and a fourth lens element. The first lens element with negative refractive power has a concave image-side surface. The second lens element with refractive power has a convex image-side surface. The third lens element with positive refractive power has a convex image-side surface. The fourth lens element with negative refractive power has a concave object-side surface and a convex image-side surface, wherein both of the object-side surface and the image-side surface of the fourth lens element are aspheric. Each of the first through fourth lens elements is a single and non-cemented lens element with refractive power. When a curvature radius of an object-side surface of the third lens element is R5, a curvature radius of the image-side surface of the third lens element is R6, a curvature radius of the object-side surface of the fourth lens element is R7, an axial distance between the first lens element and the second lens element is T12, and a focal length of the wide-angle image capturing lens assembly is f, the following relationships are satisfied:

$$0.6 < (R5+R6)/(R5-R6) < 2.0;$$

$$0.30 < R7/R6 < 0.90; \text{ and}$$

$$0.80 < T12/f < 2.5.$$

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

FIG. 1 is a schematic view of a wide-angle image capturing lens assembly according to the 1st embodiment of the present disclosure;

FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 1st embodiment;

FIG. 3 is a schematic view of a wide-angle image capturing lens assembly according to the 2nd embodiment of the present disclosure;

FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 2nd embodiment;

FIG. 5 is a schematic view of a wide-angle image capturing lens assembly according to the 3rd embodiment of the present disclosure;

FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 3rd embodiment;

FIG. 7 is a schematic view of a wide-angle image capturing lens assembly according to the 4th embodiment of the present disclosure;

FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 4th embodiment;

FIG. 9 is a schematic view of a wide-angle image capturing lens assembly according to the 5th embodiment of the present disclosure;

FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 5th embodiment;

FIG. 11 is a schematic view of a wide-angle image capturing lens assembly according to the 6th embodiment of the present disclosure; and

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 6th embodiment.

DETAILED DESCRIPTION

A wide-angle image capturing lens assembly includes four lens elements with refractive power, in order from an object side to an image side, a first lens element, a second lens element, a third lens element and a fourth lens element. Each of the first through fourth lens elements is a single and non-cemented lens element with refractive power. That is, any two lens elements adjacent to each other are not cemented, and there is an air space between the two lens elements. Moreover, the manufacturing process of the cemented lenses is more complex than the non-cemented lenses. In particular, a second surface of one lens and a first surface of the following lens need to have accurate curvature to ensure these two lens elements will be highly cemented. However, during the cementing process, those two lens elements might not be highly cemented due to decentration and it is thereby not favorable for achieving better image quality of the wide-angle image capturing lens assembly. Therefore, the wide-angle image capturing lens assembly of the present disclosure provides four non-cemented lens elements for improving the problem generated by the cemented lens elements.

The first lens element with negative refractive power can have a convex object-side surface and has a concave image-side surface, so that the field of view of the wide-angle image capturing lens assembly can be enlarged.

The second lens element with positive refractive power can have a concave object-side surface and has a convex image-side surface. Therefore, it is favorable for providing the wide-angle image capturing lens assembly with the positive refractive power as it needs to be and correcting the astigmatism.

The third lens element with positive refractive power has a convex image-side surface. Therefore, it is favorable for reducing the sensitivity and the spherical aberration of the wide-angle image capturing lens assembly.

The fourth lens element with negative refractive power has a concave object-side surface and a convex image-side surface. Therefore, it is favorable for balancing the arrangement of the negative refractive power and correcting the astigmatism of the wide-angle image capturing lens assembly.

When a curvature radius of an object-side surface of the third lens element is $R5$, and a curvature radius of the image-side surface of the third lens element is $R6$, the following relationship is satisfied: $0.6 < (R5 + R6) / (R5 - R6) < 2.0$. Therefore, it is favorable for reducing the spherical aberration and correcting the astigmatism. Preferably, the following relationship is satisfied: $0.8 < (R5 + R6) / (R5 - R6) < 1.5$. More preferably, the following relationship is satisfied: $0.85 < (R5 + R6) / (R5 - R6) < 1.3$.

When the curvature radius of the image-side surface of the third lens element is $R6$, and a curvature radius of the object-side surface of the fourth lens element is $R7$, the following relationship is satisfied: $0.30 < R7 / R6 < 0.90$. Therefore, it is

favorable for correcting the aberration so as to improve image quality. Preferably, the following relationship is satisfied: $0.30 < R7 / R6 < 0.74$.

The wide-angle image capturing lens assembly can further comprise a stop, such as an aperture stop, which locates between the second lens element and the third lens element. Therefore, it is favorable for enlarging the field of view of the wide-angle image capturing lens assembly and thereby provides a wider field of view for the same.

When an axial distance between the first lens element and the second lens element is $T12$, and a focal length of the wide-angle image capturing lens assembly is f , the following relationship is satisfied: $0.80 < T12 / f < 2.5$. Therefore, it is favorable for assembling the lens elements and maintaining a proper total track length of the wide-angle image capturing lens assembly. Preferably, the following relationship is satisfied: $1.2 < T12 / f < 2.0$.

When the focal length of the wide-angle image capturing lens assembly is f , and a focal length of the third lens element is $f3$, the following relationship is satisfied: $0.9 < f / f3 < 1.7$. Therefore, it is favorable for reducing the sensitivity of the wide-angle image capturing lens assembly.

When the focal length of the wide-angle image capturing lens assembly is f , and a focal length of the second lens element is $f2$, the following relationship is satisfied: $|f / f2| < 0.40$. Therefore, it is favorable for reducing the aberration and correcting the distortion by compensating with the first lens element.

When an Abbe number of the second lens element is $V2$, an Abbe number of the third lens element is $V3$, and an Abbe number of the fourth lens element is $V4$, the following relationship is satisfied: $0.7 < (V2 + V4) / V3 < 1.2$. Therefore, it is favorable for correcting the chromatic aberration of the wide-angle image capturing lens assembly.

When a maximal field of view of the wide-angle image capturing lens assembly is FOV , the following relationship is satisfied: $120 \text{ degrees} < FOV < 200 \text{ degrees}$. Therefore, the wide-angle image capturing lens assembly can have a larger field of view for increasing image capturing range.

When the curvature radius of the object-side surface of the fourth lens element is $R7$, and a curvature radius of the image-side surface of the fourth lens element is $R8$, the following relationship is satisfied: $-0.50 < (R7 - R8) / (R7 + R8) < -0.10$. Therefore, it is favorable for correcting the astigmatism. Preferably, the following relationship is satisfied: $-0.40 < (R7 - R8) / (R7 + R8) < -0.20$.

When the focal length of the third lens element is $f3$, and a focal length of the fourth lens element is $f4$, the following relationship is satisfied: $-0.6 < f3 / f4 < -0.3$. Therefore, it is favorable for balancing the arrangement of the refractive power of the wide-angle image capturing lens assembly and reducing the sensitivity thereof.

When a composite focal length of the lens elements with refractive power disposed between an object and the stop is ffs , and a composite focal length of the lens elements with refractive power disposed between the stop and an image plane is frs , the following relationship is satisfied: $0 < frs / ffs < 0.25$. Therefore, it is favorable for balancing the refractive power of the lens elements configured on the object side and the image side of the stop so as to enlarge the field of view of the wide-angle image capturing lens assembly and thereby provides a wider field of view for the same. The refractive power of the lens elements configured on the object side of the stop as mentioned above refers to the refractive power of the lens elements disposed between the object and the stop. The refractive power of the lens elements configured on the image

side of the stop as mentioned above refers to the refractive power of the lens elements disposed between the stop and the image plane.

When an axial distance between the third lens element and the fourth lens element is T34, and the focal length of the wide-angle image capturing lens assembly is f, the following relationship is satisfied: $0.05 < T34/f < 0.20$. Therefore, it is favorable for assembling the lens elements so as to increase the manufacturing yield rate of the wide-angle image capturing lens assembly.

When the composite focal length of the lens elements with refractive power disposed between the object and the stop is ffs, and the focal length of the wide-angle image capturing lens assembly is f, the following relationship is satisfied: $0 < f/ffs$. Therefore, the total track length will not be too long through a proper arrangement of the refractive power of the lens elements configured on the object side of the stop.

According to the wide-angle image capturing lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of glass material, the distribution of the refractive power of the wide-angle image capturing lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be aspheric, so that it is easier to make the surfaces into non-spherical shapes. As a result, more controllable variables are obtained, and the aberration is reduced, as well as the number of required lens elements can be reduced while constructing an optical system. Therefore, the total track length of the wide-angle image capturing lens assembly can also be reduced.

According to the wide-angle image capturing lens assembly of the present disclosure, each of an object-side surface and an image-side surface of every lens element has a paraxial region and a peripheral region. The paraxial region refers to the region of the surface where light rays travel close to an optical axis and the peripheral region refers to the region of the surface where light rays travel away from the optical axis. Particularly, when a lens element has a convex surface, it indicates that the surface is convex at the paraxial region; and when the lens element has a concave surface, it indicates that the surface is concave at the paraxial region.

According to the wide-angle image capturing lens assembly of the present disclosure, the wide-angle image capturing lens assembly can include at least one stop, such as an aperture stop, a glare stop, or a field stop, etc. Said glare stop or said field stop is allocated for reducing stray light while retaining high image quality. Furthermore, an aperture stop can be configured as a front stop or a middle stop. A front stop which can be disposed between the object and the first lens element provides a longer distance from an exit pupil of the system to the image plane and thereby the generated telecentric effect improves the image-sensing efficiency of an image sensor. A middle stop which can be disposed between the first lens element and the image plane is favorable for enlarging the field of view of the system and thereby provides a wider field of view for the same.

According to the wide-angle image capturing lens assembly of the present disclosure, the wide-angle image capturing lens assembly is featured with good correction ability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices and tablets.

According to the above description of the present disclosure, the following 1st-6th specific embodiments are provided for further explanation.

FIG. 1 is a schematic view of a wide-angle image capturing lens assembly according to the 1st embodiment of the present disclosure. FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 1st embodiment. In FIG. 1, the wide-angle image capturing lens assembly includes four lens elements with refractive power, in order from an object side to an image side, a first lens element 110, a second lens element 120, an aperture stop 100, a third lens element 130, a fourth lens element 140, an IR-cut filter 160, a cover glass 170 and an image plane 150.

The first lens element 110 with negative refractive power has a convex object-side surface 111 and a concave image-side surface 112. The first lens element 110 is made of glass material.

The second lens element 120 with positive refractive power has a concave object-side surface 121 and a convex image-side surface 122. The second lens element 120 is made of plastic material and has the object-side surface 121 and the image-side surface 122 being both aspheric.

The third lens element 130 with positive refractive power has a convex object-side surface 131 and a convex image-side surface 132. The third lens element 130 is made of plastic material and has the object-side surface 131 and the image-side surface 132 being both aspheric.

The fourth lens element 140 with negative refractive power has a concave object-side surface 141 and a convex image-side surface 142. The fourth lens element 140 is made of plastic material and has the object-side surface 141 and the image-side surface 142 being both aspheric.

Each of the first through fourth lens elements (110-140) is a single and non-cemented lens element with refractive power. The IR-cut filter 160 and the cover glass 170 are made of glass materials which are in order located between the fourth lens element 140 and the image plane 150, and will not affect a focal length of the wide-angle image capturing lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the wide-angle image capturing lens assembly is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y^i),$$

wherein,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the distance from the point on the curve of the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

Ai is the i-th aspheric coefficient.

In the wide-angle image capturing lens assembly according to the 1st embodiment, when the focal length of the wide-angle image capturing lens assembly is f, an f-number of the wide-angle image capturing lens assembly is Fno, and half of a maximal field of view of the wide-angle image capturing lens assembly is HFOV, these parameters have the following values: f=1.42 mm; Fno=2.84; and HFOV=82.6 degrees.

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In the wide-angle image capturing lens assembly according to the 1st embodiment, when an Abbe number of the second lens element **120** is V2, an Abbe number of the third lens element **130** is V3, and an Abbe number of the fourth lens element **140** is V4, the following relationship is satisfied: $(V2+V4)/V3=0.83$.

In the wide-angle image capturing lens assembly according to the 1st embodiment, when an axial distance between the first lens element **110** and the second lens element **120** is T12, an axial distance between the third lens element **130** and the fourth lens element **140** is T34, and the focal length of the wide-angle image capturing lens assembly is f, the following relationships are satisfied: $T12/f=1.54$; and $T34/f=0.12$.

In the wide-angle image capturing lens assembly according to the 1st embodiment, when a curvature radius of the object-side surface **131** of the third lens element **130** is R5, a curvature radius of the image-side surface **132** of the third lens element **130** is R6, a curvature radius of the object-side surface **141** of the fourth lens element **140** is R7, and a curvature radius of the image-side surface **142** of the fourth lens element **140** is R8, the following relationships are satisfied: $R7/R6=0.62$; $(R5+R6)/(R5-R6)=0.88$; and $(R7-R8)/(R7+R8)=-0.27$.

In the wide-angle image capturing lens assembly according to the 1st embodiment, when the focal length of the wide-angle image capturing lens assembly is f, a focal length of the second lens element **120** is f2, a focal length of the third lens element **130** is f3, and a focal length of the fourth lens element **140** is f4, the following relationships are satisfied: $f/f2=0.20$; $f/f3=1.20$; and $f3/f4=-0.43$.

In the wide-angle image capturing lens assembly according to the 1st embodiment, when the focal length of the wide-angle image capturing lens assembly is f, a composite focal length of the lens elements with refractive power disposed between an object and the stop **100** is ffs, and a composite focal length of the lens elements with refractive power disposed between the stop **100** and the image plane **150** is frs, the following relationships are satisfied: $f/ffs=0.01$; and $f/frs=0.03$.

In the wide-angle image capturing lens assembly according to the 1st embodiment, when the maximal field of view of the wide-angle image capturing lens assembly is FOV, the following relationship is satisfied: $FOV=165.2$ degrees.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment						
f = 1.42 mm, Fno = 2.84, HFOV = 82.6 deg.						
Surface #		Curvature Radius	Thickness	Material	Abbe #	Focal Length
0	Object	Plano	Infinity			
1	Lens 1	16.215	1.000	Glass	1.729	54.5
2		2.500	2.190			
3	Lens 2	-4.580 ASP	1.159	Plastic	1.640	23.3
4		-2.533 ASP	0.664			
5	Ape. Stop	Plano	0.421			
6	Lens 3	10.165 ASP	1.265	Plastic	1.544	55.9
7		-0.661 ASP	0.170			
8	Lens 4	-0.409 ASP	0.350	Plastic	1.640	23.3
9		-0.708 ASP	0.300			

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TABLE 1-continued

1st Embodiment						
f = 1.42 mm, Fno = 2.84, HFOV = 82.6 deg.						
Surface #		Curvature Radius	Thickness	Material	Abbe #	Focal Length
10	IR-cut filter	Plano	0.300	Glass	1.517	64.2
11		Plano	0.300			
12	Cover glass	Plano	0.800	Glass	1.517	64.2
13		Plano	1.005			
14	Image	Plano	—			

Note:
Reference wavelength (d-line) is 587.6 nm.

TABLE 2

Aspheric Coefficients			
Surface #			
	3	4	6
k =	3.5691E+00	-8.8090E+00	-1.0000E+00
A4 =	1.7495E-02	-1.7980E-02	-3.4380E-02
A6 =	-4.0283E-03	1.2812E-03	-5.8189E-02
A8 =	4.8859E-04	2.0468E-04	8.7163E-02
A10 =	1.7863E-05	-7.2716E-05	-1.6733E-01
Surface #			
	7	8	9
k =	-1.4171E+00	-1.9010E+00	-2.9655E+00
A4 =	2.3023E-01	-2.4873E-01	-2.0204E-01
A6 =	-7.4793E-01	1.2700E-01	2.3778E-01
A8 =	6.4012E-01	-2.7495E-02	-1.1258E-01
A10 =	-2.2439E-01	4.0972E-02	2.7122E-02

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-14 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A1-A16 represent the aspheric coefficients ranging from the 1st order to the 16th order. This information related to Table 1 and Table 2 applies also to the Tables for the remaining embodiments, and so an explanation in this regard will not be provided again.

2nd Embodiment

FIG. 3 is a schematic view of a wide-angle image capturing lens assembly according to the 2nd embodiment of the present disclosure. FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 2nd embodiment. In FIG. 3, the wide-angle image capturing lens assembly includes, in order from an object side to an image side, a first lens element **210**, a second lens element **220**, an aperture stop **200**, a third lens element **230**, a fourth lens element **240**, an IR-cut filter **260**, a cover glass **270** and an image plane **250**.

The first lens element **210** with negative refractive power has a convex object-side surface **211** and a concave image-side surface **212**. The first lens element **210** is made of glass material.

The second lens element **220** with positive refractive power has a concave object-side surface **221** and a convex image-

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side surface **222**. The second lens element **220** is made of plastic material and has the object-side surface **221** and the image-side surface **222** being both aspheric.

The third lens element **230** with positive refractive power has a concave object-side surface **231** and a convex image-side surface **232**. The third lens element **230** is made of plastic material and has the object-side surface **231** and the image-side surface **232** being both aspheric.

The fourth lens element **240** with negative refractive power has a concave object-side surface **241** and a convex image-side surface **242**. The fourth lens element **240** is made of plastic material and has the object-side surface **241** and the image-side surface **242** being both aspheric.

Each of the first through fourth lens elements (**210-240**) is a single and non-cemented lens element with refractive power. The IR-cut filter **260** and the cover glass **270** are made of glass materials which are in order located between the fourth lens element **240** and the image plane **250**, and will not affect a focal length of the wide-angle image capturing lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment							
f = 1.39 mm, Fno = 2.84, HFOV = 82.6 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	13.251	0.800	Glass	1.729	54.5	-4.36
2		2.500	2.110				
3	Lens 2	-4.895	ASP	Plastic	1.640	23.3	7.01
4		-2.620	ASP				
5	Ape. Stop	Plano	0.399				
6	Lens 3	-100.000	ASP	Plastic	1.544	55.9	1.10
7		-0.600	ASP				
8	Lens 4	-0.387	ASP	Plastic	1.640	23.3	-2.57
9		-0.684	ASP				
10	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—
11		Plano	0.300				
12	Cover glass	Plano	0.800	Glass	1.517	64.2	—
13		Plano	1.131				
14	Image	Plano	—				

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 4

Aspheric Coefficients			
Surface #			
	3	4	6
k =	3.3693E+00	-9.5515E+00	-1.0000E+00
A4 =	1.7057E-02	-2.5388E-02	-5.2829E-02
A6 =	-8.5736E-03	5.1797E-03	-1.9298E-01
A8 =	3.0941E-03	-8.1823E-04	3.4136E-01
A10 =	-7.0590E-04	2.6398E-05	-7.5057E-01
A12 =	7.2629E-05	1.0318E-05	—

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TABLE 4-continued

Aspheric Coefficients			
Surface #			
	7	8	9
k =	-1.7738E+00	-1.8790E+00	-2.7744E+00
A4 =	1.3240E-01	-2.2375E-01	-1.8772E-01
A6 =	-7.6413E-01	4.9736E-01	2.9466E-01
A8 =	6.8945E-01	-1.5702E+00	-2.7031E-01
A10 =	-2.8401E-01	2.9623E+00	1.8831E-01
A12 =	—	-3.1916E+00	-7.8882E-02
A14 =	—	2.0415E+00	1.8350E-02
A16 =	—	-5.6589E-01	-1.0182E-03

In the wide-angle image capturing lens assembly according to the 2nd embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment. Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following relationships:

f [mm]	1.39	(R7 - R8)/(R7 + R8)	-0.28
Fno	2.84	f/f2	0.20
HFOV [deg.]	82.6	f/f3	1.26
(V2 + V4)/V3	0.83	f3/f4	-0.43
T12/f	1.52	f/ffs	0.04
T34/f	0.11	f/fs/ffs	0.08
R7/R6	0.64	FOV [deg.]	165.2
(R5 + R6)/(R5 - R6)	1.01		

3rd Embodiment

FIG. 5 is a schematic view of a wide-angle image capturing lens assembly according to the 3rd embodiment of the present disclosure. FIG. 6 shows spherical aberration curves, astig-

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matic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 3rd embodiment. In FIG. 5, the wide-angle image capturing lens assembly includes, in order from an object side to an image side, a first lens element **310**, a second lens element **320**, an aperture stop **300**, a third lens element **330**, a fourth lens element **340**, an IR-cut filter **360**, a cover glass **370** and an image plane **350**.

The first lens element **310** with negative refractive power has a convex object-side surface **311** and a concave image-side surface **312**. The first lens element **310** is made of plastic material and has the object-side surface **311** and the image-side surface **312** being both aspheric.

The second lens element **320** with positive refractive power has a concave object-side surface **321** and a convex image-side surface **322**. The second lens element **320** is made of plastic material and has the object-side surface **321** and the image-side surface **322** being both aspheric.

The third lens element **330** with positive refractive power has a convex object-side surface **331** and a convex image-side surface **332**. The third lens element **330** is made of plastic material and has the object-side surface **331** and the image-side surface **332** being both aspheric.

The fourth lens element **340** with negative refractive power has a concave object-side surface **341** and a convex image-side surface **342**. The fourth lens element **340** is made of plastic material and has the object-side surface **341** and the image-side surface **342** being both aspheric.

Each of the first through fourth lens elements (**310-340**) is a single and non-cemented lens element with refractive power. The IR-cut filter **360** and the cover glass **370** are made of glass materials which are in order located between the fourth lens element **340** and the image plane **350**, and will not affect a focal length of the wide-angle image capturing lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

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TABLE 5

3rd Embodiment								
f = 1.61 mm, Fno = 2.84, HFOV = 75.0 deg.								
Sur- face #		Curvature Radius	Thick- ness	Mate- rial	Abbe Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	69.003 ASP	0.600	Plas- tic	1.544	55.9	-4.98	
2		2.600 ASP	2.220					
3	Lens 2	-5.232 ASP	1.188	Plas- tic	1.640	23.3	7.77	
4		-2.776 ASP	1.103					
5	Ape. Stop	Plano	0.369					
6	Lens 3	5.277 ASP	1.191	Plas- tic	1.544	55.9	1.24	
7		-0.713 ASP	0.200					
8	Lens 4	-0.382 ASP	0.350	Plas- tic	1.640	23.3	-2.72	
9		-0.664 ASP	0.300					
10	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
11		Plano	0.300					
12	Cover glass	Plano	0.800	Glass	1.517	64.2	—	
13		Plano	0.982					
14	Image	Plano	—					

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 6

Aspheric Coefficients				
	Surface #			
	1	2	3	4
k =	-5.0000E+01	-6.2198E-01	3.5540E+00	-9.2045E+00
A4 =	-2.5440E-04	-2.8709E-03	1.7069E-02	1.9339E-02
A6 =	3.9346E-06	3.6891E-03	-7.9787E-03	4.9563E-03
A8 =	4.4108E-07	-4.6220E-04	3.1570E-03	-1.1436E-03
A10 =	1.4086E-09	3.1318E-05	-7.5680E-04	6.2147E-05
A12 =	—	—	7.2629E-05	1.0318E-05
	Surface #			
	6	7	8	9
k =	-1.0000E+00	-1.4764E+00	-1.9147E+00	-2.9088E+00
A4 =	-4.5153E-02	1.6183E-01	-3.2682E-01	-1.3886E-01
A6 =	2.7138E-02	-7.5085E-01	5.2483E-01	2.7283E-01
A8 =	-8.6859E-02	6.6108E-01	-1.5241E+00	-2.7466E-01
A10 =	-1.0626E-01	-2.1608E-01	2.9734E+00	1.9094E-01
A12 =	—	—	-3.2252E+00	-7.7464E-02
A14 =	—	—	1.9936E+00	1.7866E-02
A16 =	—	—	-5.2673E-01	-1.7491E-03

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In the wide-angle image capturing lens assembly according to the 3rd embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment. Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following relationships:

f [mm]	1.61	$(R7 - R8)/(R7 + R8)$	-0.27
Fno	2.84	$ f/f2 $	0.21
HFOV [deg.]	75.0	$f/f3$	1.30
$(V2 + V4)/V3$	0.83	$f3/f4$	-0.46
T12/f	1.38	$f/f5$	0.03
T34/f	0.12	$f5/f6$	0.05
R7/R6	0.54	FOV [deg]	150.0
$(R5 + R6)/(R5 - R6)$	0.76		

4th Embodiment

FIG. 7 is a schematic view of a wide-angle image capturing lens assembly according to the 4th embodiment of the present disclosure. FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 4th embodiment. In FIG. 7, the wide-angle image capturing lens assembly includes, in order from an object side to an image side, a first lens element 410, a second lens element 420, an aperture stop 400, a third lens element 430, a fourth lens element 440, an IR-cut filter 460, a cover glass 470 and an image plane 450.

The first lens element 410 with negative refractive power has a concave object-side surface 411 and a concave image-side surface 412. The first lens element 410 is made of glass material and has the object-side surface 411 and the image-side surface 412 being both aspheric.

The second lens element 420 with positive refractive power has a concave object-side surface 421 and a convex image-side surface 422. The second lens element 420 is made of plastic material and has the object-side surface 421 and the image-side surface 422 being both aspheric.

The third lens element 430 with positive refractive power has a concave object-side surface 431 and a convex image-side surface 432. The third lens element 430 is made of plastic material and has the object-side surface 431 and the image-side surface 432 being both aspheric.

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The fourth lens element 440 with negative refractive power has a concave object-side surface 441 and a convex image-side surface 442. The fourth lens element 440 is made of plastic material and has the object-side surface 441 and the image-side surface 442 being both aspheric.

Each of the first through fourth lens elements (410-440) is a single and non-cemented lens element with refractive power. The IR-cut filter 460 and the cover glass 470 are made of glass materials which are in order located between the fourth lens element 440 and the image plane 450, and will not affect a focal length of the wide-angle image capturing lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment						
f = 1.60 mm, Fno = 2.70, HFOV = 70.0 deg.						
Surface #		Curvature Radius	Thickness	Material	Abbe Index	Focal Length
0	Object	Plano	Infinity			
1	Lens 1	-7.480 ASP	0.800	Glass	1.542	62.9
2		3.044 ASP	2.002			-3.88
3	Lens 2	-97.671 ASP	1.186	Plastic	1.639	23.5
4		-3.887 ASP	0.977			6.30
5	Ape. Stop	Plano	0.235			
6	Lens 3	-13.752 ASP	1.189	Plastic	1.544	55.9
7		-0.589 ASP	0.104			1.10
8	Lens 4	-0.424 ASP	0.350	Plastic	1.640	23.3
9		-0.780 ASP	0.300			-2.36
10	IR-cut filter	Plano	0.300	Glass	1.517	64.2
11		Plano	0.300			—
12	Cover glass	Plano	0.400	Glass	1.517	64.2
13		Plano	1.774			—
14	Image	Plano	—			

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 8

Aspheric Coefficients				
	Surface #			
	1	2	3	4
k =	-5.0000E+01	1.5657E-02	5.0000E+01	-4.1324E+00
A4 =	4.2796E-06	-8.5954E-03	-1.3322E-02	4.1672E-03
A6 =	—	-6.1014E-04	1.6791E-03	-3.5170E-03
A8 =	—	1.5342E-05	—	1.2028E-03
A10 =	—	—	—	-1.7725E-04
A12 =	—	—	—	1.0318E-05
	Surface #			
	6	7	8	9
k =	-1.0000E+00	-2.2054E+00	-2.2564E+00	-3.6171E+00
A4 =	-8.5790E-02	1.5858E-01	-2.8536E-02	-1.2541E-01
A6 =	2.8649E-01	-8.0174E-01	8.5290E-02	2.0103E-01
A8 =	-1.9262E+00	3.5116E-01	-1.5257E+00	-2.6410E-01
A10 =	2.3136E+00	3.6891E-02	2.9914E+00	2.2305E-01
A12 =	—	—	-3.1372E+00	-8.2617E-02

TABLE 8-continued

Aspheric Coefficients				
A14 =	—	—	2.2882E+00	6.1130E-05
A16 =	—	—	-7.9551E-01	5.3773E-03

In the wide-angle image capturing lens assembly according to the 4th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment. Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following relationships:

f [mm]	1.60	$(R7 - R8)/(R7 + R8)$	-0.30
Fno	2.70	$ f/f2 $	0.25
HFOV [deg.]	70.0	$f/f3$	1.46
$(V2 + V4)/V3$	0.84	$f3/f4$	-0.47
T12/f	1.25	$f/f5$	0.03
T34/f	0.07	$f1s/f5s$	0.06
R7/R6	0.72	FOV [deg.]	140.0
$(R5 + R6)/(R5 - R6)$	1.09		

5th Embodiment

FIG. 9 is a schematic view of a wide-angle image capturing lens assembly according to the 5th embodiment of the present disclosure. FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 5th embodiment. In FIG. 9, the wide-angle image capturing lens assembly includes, in order from an object side to an image side, a first lens element **510**, a second lens element **520**, an aperture stop **500**, a third lens element **530**, a fourth lens element **540**, an IR-cut filter **560**, a cover glass **570** and an image plane **550**.

The first lens element **510** with negative refractive power has a convex object-side surface **511** and a concave image-side surface **512**. The first lens element **510** is made of glass material and has the object-side surface **511** and the image-side surface **512** being both aspheric.

The second lens element **520** with negative refractive power has a concave object-side surface **521** and a convex image-side surface **522**. The second lens element **520** is made of plastic material and has the object-side surface **521** and the image-side surface **522** being both aspheric.

The third lens element **530** with positive refractive power has a concave object-side surface **531** and a convex image-side surface **532**. The third lens element **530** is made of plastic material and has the object-side surface **531** and the image-side surface **532** being both aspheric.

The fourth lens element **540** with negative refractive power has a concave object-side surface **541** and a convex image-side surface **542**. The fourth lens element **540** is made of plastic material and has the object-side surface **541** and the image-side surface **542** being both aspheric.

Each of the first through fourth lens elements (**510-540**) is a single and non-cemented lens element with refractive power. The IR-cut filter **560** and the cover glass **570** are made of glass materials which are in order located between the fourth lens element **540** and the image plane **550**, and will not affect a focal length of the wide-angle image capturing lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment							
f = 1.22 mm, Fno = 2.84, HFOV = 75.0 deg.							
Surface #		Curvature Radius	Thick-ness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	30.057 ASP	0.600	Glass	1.542	62.9	-6.27
2		3.036 ASP	2.309				
3	Lens 2	-3.332 ASP	0.991	Plastic	1.640	23.3	-40.82
4		-4.263 ASP	1.045				
5	Ape. Stop	Plano	0.136				
6	Lens 3	-100.000 ASP	1.519	Plastic	1.544	55.9	0.87
7		-0.473 ASP	0.098				
8	Lens 4	-0.345 ASP	0.350	Plastic	1.640	23.3	-2.10
9		-0.649 ASP	0.300				
10	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—
11		Plano	0.300				
12	Cover glass	Plano	0.800	Glass	1.517	64.2	—
13		Plano	1.160				
14	Image	Plano	—				

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 10

Aspheric Coefficients				
Surface #				
	1	2	3	4
k =	2.6313E+01	-1.0647E-01	1.0637E+00	-6.8784E+00
A4 =	1.9626E-03	-1.1069E-02	1.9360E-02	-1.5574E-02
A6 =	-2.9898E-05	1.2225E-03	-5.0028E-03	3.8681E-03
A8 =	-9.9801E-07	-1.2087E-04	2.4656E-03	-1.1340E-03
A10 =	—	3.1014E-06	-6.4289E-04	1.1689E-04
A12 =	—	—	7.2629E-05	1.0318E-05
Surface #				
	6	7	8	9
k =	-1.0000E+00	-1.7439E+00	-1.6798E+00	-2.1721E+00
A4 =	-1.3652E-01	1.4043E-01	1.0359E-01	-6.7087E-02
A6 =	2.5762E-01	-7.1719E-01	1.1458E-01	2.2324E-01
A8 =	-8.8606E+00	2.5050E-01	-1.5703E+00	-3.0671E-01
A10 =	2.3705E+01	1.0348E-01	3.0521E+00	2.6591E-01
A12 =	—	—	-2.9217E+00	-9.3498E-02
A14 =	—	—	2.0516E+00	-6.9251E-03
A16 =	—	—	-7.7231E-01	9.5849E-03

In the wide-angle image capturing lens assembly according to the 5th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment. Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following relationships:

f [mm]	1.22	(R7 - R8)/(R7 + R8)	-0.31
Fno	2.84	f/f2	0.03
HFOV [deg.]	75.0	f/f3	1.41
(V2 + V4)/V3	0.83	f3/f4	-0.41
T12/f	1.89	f/f5	-0.22
T34/f	0.08	f5/f6	-0.37
R7/R6	0.73	FOV [deg.]	150.0
(R5 + R6)/(R5 - R6)	1.01		

6th Embodiment

FIG. 11 is a schematic view of a wide-angle image capturing lens assembly according to the 6th embodiment of the present disclosure. FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the wide-angle image capturing lens assembly according to the 6th embodiment. In FIG. 11, the wide-angle image capturing lens assembly includes, in order from an object side to an image side, a first lens element 610, a second lens element 620, an aperture stop 600, a third lens element 630, a fourth lens element 640, an IR-cut filter 660, a cover glass 670 and an image plane 650.

The first lens element 610 with negative refractive power has a convex object-side surface 611 and a concave image-side surface 612. The first lens element 610 is made of plastic material and has the object-side surface 611 being aspheric.

The second lens element 620 with positive refractive power has a convex object-side surface 621 and a convex image-side surface 622. The second lens element 620 is made of plastic material and has the image-side surface 622 being aspheric.

The third lens element 630 with positive refractive power has a concave object-side surface 631 and a convex image-side surface 632. The third lens element 630 is made of plastic

material and has the object-side surface 631 and the image-side surface 632 being both aspheric.

The fourth lens element 640 with negative refractive power has a concave object-side surface 641 and a convex image-side surface 642. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being both aspheric.

Each of the first through fourth lens elements (610-640) is a single and non-cemented lens element with refractive power. The IR-cut filter 660 and the cover glass 670 are made of glass materials which are in order located between the fourth lens element 640 and the image plane 650, and will not affect a focal length of the wide-angle image capturing lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment							
f = 1.61 mm, Fno = 2.50, HFOV = 70.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	100.000 ASP	0.800	Plastic	1.535	55.7	-4.55
2		2.370	2.518				
3	Lens 2	12.409	0.486	Plastic	1.639	23.5	5.52
4		-4.858 ASP	0.419				
5	Ape. Stop	Plano	0.232				
6	Lens 3	-13.752 ASP	1.244	Plastic	1.544	55.9	1.12
7		-0.600 ASP	0.163				
8	Lens 4	-0.388 ASP	0.440	Plastic	1.640	23.3	-2.96
9		-0.705 ASP	0.300				
10	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—
11		Plano	0.300				
12	Cover glass	Plano	0.400	Glass	1.517	64.2	—

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TABLE 11-continued

6th Embodiment f = 1.61 mm, Fno = 2.50, HFOV = 70.0 deg.					
Sur- face #	Curvature Radius	Thick- ness	Mate- rial	Index	Abbe #
13	Plano	1.399			
14 Image	Plano	—			

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 12

Aspheric Coefficients			
	Surface #		
	1	4	6
k =	-5.0000E+01	-4.5491E+00	-1.0000E+00
A4 =	7.3789E-05	5.4261E-03	-1.1514E-01
A6 =	—	-3.2241E-03	4.3795E-01
A8 =	—	1.2151E-03	-2.1725E+00
A10 =	—	-1.8961E-04	2.5705E+00
A12 =	—	1.0318E-05	
	Surface #		
	7	8	9
k =	-1.2203E+00	-1.7325E+00	-2.4138E+00
A4 =	4.7481E-01	-2.2055E-02	-1.3447E-01
A6 =	-9.8833E-01	9.2008E-02	1.9980E-01
A8 =	4.1302E-01	-1.5219E+00	-2.6278E-01
A10 =	2.3294E-02	2.9945E+00	2.2444E-01
A12 =	—	-3.1336E+00	-8.1933E-02
A14 =	—	2.2922E+00	2.9696E-04
A16 =	—	-7.9520E-01	5.2645E-03

In the wide-angle image capturing lens assembly according to the 6th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment. Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following relationships:

f [mm]	1.61	(R7 - R8)/(R7 + R8)	-0.29
Fno	2.50	f/f2	0.29
HFOV [deg.]	70.0	f/f3	1.45
(V2 + V4)/V3	0.84	f3/f4	-0.38
T12/f	1.56	f/f5	0.11
T34/f	0.10	f5/f6	0.18
R7/R6	0.65	FOV [deg.]	140.0
(R5 + R6)/(R5 - R6)	1.09		

It is to be noted that TABLES 1-12 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. Therefore, any imaging lens system of the same structure is considered to be less than or equal to the scope of the present disclosure even if it uses different data. The embodiments depicted above and the appended drawings are exemplary and are not intended to limit the scope of the present disclosure.

What is claimed is:

1. A wide-angle image capturing lens assembly comprising four lens elements with refractive power, in order from an object side to an image side:

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a first lens element with negative refractive power having a concave image-side surface;

a second lens element with refractive power having a concave object-side surface and a convex image-side surface;

a third lens element with positive refractive power having a convex image-side surface; and

a fourth lens element with negative refractive power having a concave object-side surface and a convex image-side surface, wherein both of the object-side surface and the image-side surface of the fourth lens element are aspheric;

wherein each of the first through fourth lens elements is a single and non-cemented lens element with refractive power, a curvature radius of an object-side surface of the third lens element is R5, a curvature radius of the image-side surface of the third lens element is R6, a curvature radius of the object-side surface of the fourth lens element is R7, a maximal field of view of the wide-angle image capturing lens assembly is FOV, and the following relationships are satisfied:

$$0.6 < (R5 + R6) / (R5 - R6) < 2.0;$$

$$0.30 < R7 / R6 < 0.74; \text{ and}$$

$$120 \text{ degrees} < \text{FOV} < 200 \text{ degrees.}$$

2. A wide-angle image capturing lens assembly comprising, four lens elements with refractive power, in order from an object side to an image side:

a first lens element with negative refractive power having a concave image-side surface;

a second lens element with refractive power having a convex image-side surface;

a third lens element with positive refractive power having a convex image-side surface; and

a fourth lens element with negative refractive power having a concave object-side surface and a convex image-side surface, wherein both of the object-side surface and the image-side surface of the fourth lens element are aspheric;

wherein each of the first through fourth lens elements is a single and non-cemented lens element with refractive power, a curvature radius of an object-side surface of the third lens element is R5, a curvature radius of the image-side surface of the third lens element is R6, a curvature radius of the object-side surface of the fourth lens element is R7, an axial distance between the first lens element and the second lens element is T12, a focal length of the wide-angle image capturing lens assembly is f, an Abbe number of the second lens element is V2, an Abbe number of the third lens element is V3, an Abbe number of the fourth lens element is V4, and the following relationships are satisfied:

$$0.6 < (R5 + R6) / (R5 - R6) < 2.0;$$

$$0.30 < R7 / R6 < 0.90;$$

$$0.80 < T12 / f < 2.5; \text{ and}$$

$$0.7 < (V2 + V4) / V3 < 1.2.$$

3. The wide-angle image capturing lens assembly of claim 1, further comprising:

a stop located between the second lens element and the third lens element.

4. The wide-angle image capturing lens assembly of claim 1, wherein a focal length of the wide-angle image capturing lens

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assembly is f , a focal length of the second lens element is f_2 , and the following relationship is satisfied:

$$|f/f_2| < 0.40.$$

5. The wide-angle image capturing lens assembly of claim 1, wherein the second lens element has positive refractive power and at least one of the object-side surface and the image-side surface of the second lens element is aspheric.

6. The wide-angle image capturing lens assembly of claim 1, further comprising a stop, wherein a composite focal length of the lens elements with refractive power disposed between an object and the stop is ffs , a composite focal length of the lens elements with refractive power disposed between the stop and an image plane is f_{fs} , and the following relationship is satisfied:

$$0 < f_{fs}/ffs < 0.25.$$

7. The wide-angle image capturing lens assembly of claim 2, wherein the curvature radius of the object-side surface of the fourth lens element is R_7 , a curvature radius of the image-side surface of the fourth lens element is R_8 , and the following relationship is satisfied:

$$-0.50 < (R_7 - R_8)/(R_7 + R_8) < -0.10.$$

8. The wide-angle image capturing lens assembly of claim 2, wherein the curvature radius of the image-side surface of the third lens element is R_6 , the curvature radius of the object-side surface of the fourth lens element is R_7 , and the following relationship is satisfied:

$$0.30 < R_7/R_6 < 0.74.$$

9. The wide-angle image capturing lens assembly of claim 2, wherein the focal length of the wide-angle image capturing lens assembly is f , a focal length of the second lens element is f_2 , and the following relationship is satisfied:

$$|f/f_2| < 0.40.$$

10. The wide-angle image capturing lens assembly of claim 2, further comprising a stop, wherein a composite focal length of the lens elements with refractive power disposed between an object and the stop is ffs , the focal length of the wide-angle image capturing lens assembly is f , and the following relationship is satisfied:

$$0 < f/ffs.$$

11. The wide-angle image capturing lens assembly of claim 3, wherein an axial distance between the first lens element and the second lens element is T_{12} , a focal length of the wide-angle image capturing lens assembly is f , and the following relationship is satisfied:

$$0.80 < T_{12}/f < 2.5.$$

12. The wide-angle image capturing lens assembly of claim 4, wherein the curvature radius of the object-side surface of the third lens element is R_5 , the curvature radius of the

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image-side surface of the third lens element is R_6 , and the folio wring relationship is satisfied:

$$0.8 < (R_5 + R_6)/(R_5 - R_6) < 1.5.$$

13. The wide-angle image capturing lens assembly of claim 4, wherein an Abbe number of the second lens element is V_2 , an Abbe number of the third lens element is V_3 , an Abbe number of the fourth lens element is V_4 , and the following relationship is satisfied:

$$0.7 < (V_2 + V_4)/V_3 < 1.2.$$

14. The wide-angle image capturing lens assembly of claim 5, wherein the curvature radius of the object-side surface of the fourth lens element is R_7 , a curvature radius of the image-side surface of the fourth lens element is R_8 , and the following relationship is satisfied:

$$-0.40 < (R_7 - R_8)/(R_7 + R_8) < -0.20.$$

15. The wide-angle image capturing lens assembly of claim 5, wherein the first lens element has a convex object-side surface.

16. The wide-angle image capturing lens assembly of claim 7, wherein an axial distance between the third lens element and the fourth lens element is T_{34} , the focal length of the wide-angle image capturing lens assembly is f , and the following relationship is satisfied:

$$0.05 < T_{34}/f < 0.20.$$

17. The wide-angle image capturing lens assembly of claim 10, wherein the first lens element has a convex object-side surface.

18. The wide-angle image capturing lens assembly of claim 10, wherein the axial distance between the first lens element and the second lens element is T_{12} , the focal length of the wide-angle image capturing lens assembly is f , and the following relationship is satisfied:

$$1.2 < T_{12}/f < 2.0.$$

19. The wide-angle image capturing lens assembly of claim 11, wherein the curvature radius of the object-side surface of the third lens element is R_5 , the curvature radius of the image-side surface of the third lens element is R_6 , and the following relationship is satisfied:

$$0.85 < (R_5 + R_6)/(R_5 - R_6) < 1.3.$$

20. The wide-angle image capturing lens assembly of claim 11, wherein the focal length of the wide-angle image capturing lens assembly is f , a focal length of the third lens element is f_3 , and the following relationship is satisfied:

$$0.9 < f/f_3 < 1.7.$$

21. The wide-angle image capturing lens assembly of claim 15, wherein a focal length of the third lens element is f_3 , a focal length of the fourth lens element is f_4 , and the following relationship is satisfied:

$$-0.6 < f_3/f_4 < -0.3.$$

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 8,982,479 B2
APPLICATION NO. : 13/792178
DATED : March 17, 2015
INVENTOR(S) : Hsieh et al.

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims

Column 20, line 37, Claim 2 delete “a fourth lens element with negative refractive pourer”, and insert -
-a fourth lens element with negative refractive power--.

Column 22, line 2, Claim 12 delete “and the folio wring relationship is satisfied”, and insert --and the
following relationship is satisfied--.

Column 22, line 30, Claim 17 delete “the first lens element has a convex object-id surface”, and insert
--the first lens element has a convex object-side surface--.

Column 22, line 33, Claim 18 delete “...second lens element is T12 the focal length...”, and insert
--...second lens element is T12, the focal length...--.

Signed and Sealed this
Thirtieth Day of June, 2015



Michelle K. Lee
Director of the United States Patent and Trademark Office